

Bachelor in Computer Applications

Course Title: **Information Security**

Code No: Semester: **5**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Consider $p=11$ and $q=7$ in a RSA cryptosystem.i. What is a public key pair (e, n) ?ii. What is a private key pair (d, n) ?iii. What is ciphertext for $M=6$?
2. Discuss how encryption and decryption is done in the DES algorithm.
3. Define subjects, objects and access rights in access control with suitable examples. How role based access control is different from attribute based access control?
4. How online and offline dictionary attacks are done in password based authentication systems?
5. Describe the roles of relying parties, attribute providers and identity providers in Open Identity Trust Framework.
6. Define zombies, rootkits and Trojans.
7. Briefly describe the status of cyber law in Nepal.
8. Discuss various methods of risk treatment during security risk analysis.
9. What is the use of S-box in DES? Illustrate S-box operation with an example.
10. How hash value is generated by the SHA-2 hash function.
11. Write Rabin Miller Algorithm for primality testing. Test whether 341 is prime or not using the algorithm.
12. Define interception, repudiation and incapacitation with examples.